Harvesting names from unlabeled papers to augment materials named entity recognition on unseen documents

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Abstract

Materials named-entity recognizers train on few annotated papers but read new ones, where many entities are names no training sentence annotates. We harvest such names from 40,000 unlabeled sentences, keeping those that three baseline taggers confidently agree on, and add them to mention replacement, which swaps names in copied training sentences. Over 11 held-out subfields and papers of two corpora this gains 3.58 F1 points [95% CI 0.95, 6.21] over the baseline, and every MaterioMiner paper gains at both granularities. The pooled criterion was adopted mid-study, when mention replacement alone met it; the registered one, gains on MuLMS and MaterioMiner, was not met, and a third corpus shows none. Against a six-copy control the names' F1 increment, 0.80 [−0.66, 2.26] on seeds unused in selection, is not established, but on MuLMS they recall more of the unseen names in their lexicon than that control at any decoding bias (post hoc).

Introduction

Annotated materials corpora are small and come from few documents. MaterioMiner [1] annotates 4 papers; SOFC-Exp [2] 45 articles; MuLMS [3] 50 articles from 7 subfields. A named-entity tagger trained on such a corpus is meant for documents the corpus does not contain, and a random sentence split does not measure that. Our baseline is MatSciBERT [4], a BERT language model [5] pre-trained on materials text, with a conditional random field (CRF) output layer: the *CRF head* (Methods, *Systems*). Scored by micro F1 (the harmonic mean of precision and recall over entities matched exactly in span and type, in percentage points; Methods, *Evaluation protocol*), it reaches 77.49 with MaterioMiner's fine-grained types (its 27 most frequent classes; the coarse-grained level has 27 parent types, Table 1) over 5 random splits and 46.46 on a held-out paper; on MuLMS 65.72 over random splits of the same size and 48.66 on a held-out subfield; on SOFC-Exp, whose 45 articles share one topic and one annotation scheme, 86.27 against 82.79 on held-out articles. The gap is partly the names: 81% of the entities of a held-out MaterioMiner paper are *unseen*, their words annotated as no entity in any training sentence (88% on a held-out MuLMS subfield, 46% on SOFC-Exp's held-out articles), and the CRF head finds 35% of them on MaterioMiner and 39% on MuLMS, against 71% and 66% of the names it has seen. The held-out document's mix of names accounts for a third to three fifths of the recall gap, by corpus and by the order of the decomposition; the rest is lower recall on both kinds (a companion study, H.M., in preparation; Methods, *Relation to a companion study*).

Text mining has become a source of materials data [6,7]. Besides MatSciBERT, ChemDataExtractor [8] combines dictionaries, rules and a CRF tagger; Weston et al. [9] applied a materials NER model at scale; Trewartha et al. [10] and MatSci-NLP [11] benchmark materials language models; and large language models now extract structured materials information by fine-tuning [12] or prompting [13]. Potu et al. [14] report 66.74 F1 on MaterioMiner against 78.56 in distribution, a gap of about 12 points where our three-paper models lose 28–31 (companion study). Random sentence splits leak names between training and test [15], and their use is disputed [16,17]; taggers score far lower on unseen mentions [18–20], lean on name regularity and the names themselves [21,22], narrow but do not close that gap with contextual embeddings [23], and are exposed when a test set's entities are switched [24]. Seed variance and significance on small test sets are treated by Reimers and Gurevych [25], Dror et al. [26] and Bouthillier et al. [27], and Card et al. [28] show how underpowered typical NLP comparisons are, which is why Supplementary Note 9 reports what each comparison here could detect.

Mention replacement, which swaps annotated mentions for others of the same type, is one of the simple augmentations Dai and Adel [29] compared on a materials synthesis corpus [30] and a clinical one; it helped most

with small training sets, and its gains shrink and reverse as training grows [31]; most augmentations make the tagger memorise entities [32], which is why unseen entities are scored separately here. The replacements otherwise come from the training set [33], a masked language model [34], a generative model [35], related sentences [36] or lists of the target domain's names [37]; DAGA [38] and LADA [39] generate or interpolate labeled examples, and cross-domain work [40,41] changes domain, not document. We change only where the names come from, keep every gold context, and test on held-out documents of the same corpus.

Gazetteers have been bootstrapped from unlabeled text since Riloff and Jones [42] and Kozareva [43], and agreement between models as a filter on unlabeled examples goes back to co-training and tri-training [44,45]. Zafarian and Asghari [46] keep a supervised tagger's confident entities as a gazetteer feature; GEMNET [47] gates gazetteer features against the context; Song et al. [48] find no clear gain from replacing annotated entities with gazetteer entries, on a corpus whose test names are mostly seen. AutoNER [49], BOND [50] and weak supervision with labelling functions [51] train taggers from dictionaries, heuristics and unlabeled text; students learn from several teachers on unlabeled target-language text [52], and teachers' agreement selects labels in distantly supervised NER [53]. Our self-training variant follows BOND on the topic corpus, where Du et al. [54] retrieve sentences from the web. What this paper adds is the test: the lexicon's contribution isolated against a matched control and its recall against a recall-matched decoding bias, its source and settings varied, on held-out documents, each comparison fixed before its new runs. The combination improves on the baseline on held-out documents of two corpora, and, post hoc, the names recall more of the unseen names their lexicon holds than the matched control does at any decoding bias tried. The registered recall-matched test (D11) passes only with three subfields that no bias matched (+6.05 [−3.33, +15.44] on the four matched), and the combination's gain in F1 is not shown to exceed mention replacement's (D7).

This paper asks whether the unseen names can be given to the tagger without annotation. Unlabeled materials text is plentiful: a corpus of 40,000 sentences of open-access papers on a held-out MuLMS subfield's topic holds 55–81% of its unseen test names. Three CRF heads trained on the labeled sentences (the *teachers*) tag the corpus; the names all three tag alike, each tag with a probability (its *marginal*) of 0.9 or more, form a *lexicon*; and the lexicon enters training through mention replacement [29], which copies each training sentence six times with its entity mentions swapped for other mentions of the same type, drawn from the training mentions and the lexicon together (*MR6 + names*; Figure 1 and Methods, *Systems*). The model trained on the augmented data, the *student*, never trains on a teacher-labeled sentence: it sees only annotated sentences, some of whose names are swapped for harvested ones. The Methods (*Terms used throughout*) collect these and the other terms.

Every decision was fixed in a protocol committed to the repository before its results [55], with 7 to 10 training runs (*seeds*) per held-out *unit* (a MuLMS subfield, a MaterioMiner paper or a SOFC-Exp fold of articles) and a decision rule stated in advance: a system improves on its reference when the 95% interval of the mean paired difference in micro F1 over the units lies above 0. A number in square brackets is that 95% Student-t confidence interval, over the units named; a comparison specified after the results of the runs it analyses were known is labelled post hoc. Protocols are cited by date, with the time where one day has two, and Supplementary Note 1 lists each with its commit. The *names route* is the sequence of protocols that led to and tested harvested names; five of them (18 October 11:00 to 1 November) took the twelve *decisions* on MR6 + names, D1 to D12 (Table 2), beside others counted at the end of the Results. The results, with the qualifications recorded in the Methods (*Evaluation protocol*):

- MR6 + names improves on the CRF head on MaterioMiner's held-out papers at both granularities (D3, D4) and pooled over the 11 units of MuLMS and MaterioMiner (D5; Results, *The pre-registered decisions*), under a pooled criterion adopted mid-study, when plain mention replacement was already known to meet it (Methods, *Evaluation protocol*). MuLMS alone does not pass (D1).
- Most of that gain is mention replacement's. Mention replacement with three copies and no names (MR3) gains about two thirds of it on the same units (post hoc), and the pre-registered test of the names against six-copy mention replacement (MR6; D7) excludes an increment above 2.3 points but not a smaller one (Results,

Whose gain is it); macro and weighted F1 read the same way on the seeds not used in selection (the *decision seeds*; post hoc: macro F1 +0.83 [−0.01, +1.67], weighted F1 +0.89 [−0.40, +2.18]; Results, *The pre-registered decisions*).

- Neither the names' source nor the lexicon's settings changes F1 detectably (Results, *The names' source and the lexicon's settings*). On MuLMS the names raise recall on unseen names, most on the fifth of them the lexicon holds (+7.29 [+3.27, +11.30] over MR6, post hoc), by more than MR6 gains there at any decoding bias of the grid (post hoc). The registered test against MR6 decoded at a bias matching their overall recall (D11) passes, but only with three subfields that no bias matched (+6.05 [−3.33, +15.44] on the four matched), and the names are not shown to gain F1 over it (D12; Results, *What the names do*). On SOFC-Exp, where half the test names are seen, no system is shown to improve on the baseline (Results, *SOFC-Exp: no gain shown*).
- Two follow-ups were registered after D1 to D8 had been written up: the recall-matched control (D11, D12) and a test of the names on MaterioMiner's strongest system, multi-task training (one encoder with a CRF head for each of MaterioMiner's two type levels) with mention replacement, where against a six-copy matched control neither of its two decisions passes (D9, D10; Results, *The names on the strongest system*).

Four of the twelve decisions pass: three on F1, which share MaterioMiner's four papers (D3 and D4 at its two granularities, and D5, whose 11 units include them), and D11, on recall, which passes only with three subfields that no bias matched and whose outcome was chosen after the lexicon split had found the effect: it tests one alternative explanation, a uniform shift toward tagging more spans, not the effect afresh. None would survive a Bonferroni correction (Results, *The pre-registered decisions*). The last part of the Results counts what else was tried on the way.

Results

The pre-registered decisions

Table 2 lists the twelve decisions of the names route in the order they were taken, with the seeds each used; D9 and D10 belong to the follow-up of *The names on the strongest system*, D11 and D12 to that of *What the names do* (both below).

Three F1 decisions pass, and they are not three pieces of evidence: D3 and D4 are the same four papers at two granularities, and D5's eleven units contain those four papers (the fourth pass, D11, is on recall and holds only with three subfields that no bias matched; Results, *What the names do*). On MuLMS, the corpus of the pooled decision that is independent of those papers, MR6 + names fails D1 and D2, while MR6 alone clears 0 against the CRF head on the decision seeds (+2.68 [+0.13, +5.23]; on all ten +2.33 [−0.27, +4.92], covering 0; pre-specified on 29 October), with a smaller mean than MR6 + names' +3.19 but a narrower spread; with MR6 + names 6 of the 7 subfields gain (sofc by 11.9, poly loses 2.4), and the differences have a standard deviation of 4.7 points across subfields. On MaterioMiner every paper gains at both granularities, but with four units the sign-flip test cannot go below P = 0.125, so D3 and D4 rest on the t interval with 3 degrees of freedom. The pooled estimate survives leaving out any one unit (post hoc: the lowest lower bounds are +0.44 and +0.54, without semiconductors), and the other post hoc checks agree: weighting the two corpora equally gives 3.73 on either seed basis, with Welch intervals [+1.39, +6.07] and [+1.46, +6.00] (D7's is +0.90 [−0.88, +2.69]); 9 of the 11 units are positive on the executed 10-seed basis and 10 on the 7-seed basis; and the exact sign-flip test gives P = 0.011 and 0.007. Figure 2 shows every unit.

The decision rule carries no correction for the number of decisions. Under a Bonferroni bound of 0.05/12 none of the four passes would hold: their two-sided P values are 0.013 to 0.018 (0.011 for D5′; Table 2). Mention replacement's post hoc pooled gain over the CRF head (Results, *Whose gain is it*) has P = 0.004 on either seed basis, below 0.05/12, but as a post hoc comparison it belongs to no corrected family.

Other measures. The protocols of 17 and 18 October named macro F1 (every type weighted equally), weighted F1 (types weighted by their test entities) and the votes as reported beside micro F1; only mention replacement's

votes of 17 October decided. On MuLMS macro F1 varies far less across subfields than micro F1, so its interval clears 0 in 23 per-corpus comparisons where micro F1's does not, every one against the CRF head or MR3, among them D1 (+3.78 [+2.47, +5.10]); the converse occurs 8 times, 7 on MaterioMiner or SOFC-Exp (Supplementary Note 12). Against the copy-matched control, and on MuLMS alone, the secondary measures give the micro picture. On MuLMS's decision seeds macro and weighted F1 give +1.10 [−0.46, +2.65] and +1.17 [−1.09, +3.44] over MR3 (D2), +0.53 [−0.56, +1.62] and +0.84 [−1.04, +2.71] over MR6 (D7's MuLMS part), and −0.24 [−1.56, +1.08] and −0.15 [−2.06, +1.75] against the mixed corpus (D8). Pooled over D7's 11 units (post hoc), the names over MR6 give +0.83 [−0.01, +1.67] in macro and +0.89 [−0.40, +2.18] in weighted F1 on the decision seeds, and +1.12 [+0.21, +2.03] and +1.22 [−0.21, +2.65] on all ten, which add the seeds 0 to 200 on which MR6 + names was selected. Over MR3 the pooled macro-F1 increment clears 0 (+1.33 [+0.29, +2.36] on the decision seeds, +1.73 [+0.84, +2.61] on all ten; weighted F1 +1.25 [−0.29, +2.78] and +1.53 [+0.26, +2.81]), the three extra copies included (MR6 over MR3, +0.50 [−0.31, +1.30] in macro and +0.36 [−0.11, +0.82] in weighted F1). In the measures no decision used, the names' increment over the copy-matched control thus has intervals that include 0 on the decision seeds (macro F1's lower bound at −0.01, weighted F1's at −0.40), and one above 0 only in macro F1 with the selection seeds. The votes read as micro F1 does wherever they were computed: D1 +3.23 [−1.15, +7.61], D2 +0.57 [−2.22, +3.36], D3 +4.99 [+3.27, +6.72], D4 +2.14 [+0.23, +4.05], D7's parts +0.35 [−1.67, +2.36] (MuLMS) and +0.94 [−1.60, +3.49] (MaterioMiner), D8 −0.39 [−2.64, +1.86], D9 +0.19 [−2.28, +2.66] and D10 +0.63 [−1.65, +2.91]; they were not computed for the pooled comparisons or on SOFC-Exp. Supplementary Note 12 lists all 162 rows of the names route in every measure, with the votes and, where the results hold them, the per-unit values.

Whose gain is it

Table 3 gives every comparison among the four systems on the same 11 units, on both seed bases, and on MaterioMiner per granularity, with the status of each.

The +3.58 decomposes, on the decision seeds, into +2.46 from mention replacement with three copies, +0.33 from three more copies and +0.80 from the harvested names; only the first term has an interval above 0, and that post hoc (mention replacement's own pre-registered single-model decisions passed on MaterioMiner and failed on MuLMS alone, where its votes passed; Table 3 and Supplementary Note 7). The 10-seed basis gives the same picture (+2.35, +0.22, +1.02). Against MR3 the whole increment of MR6 + names, +1.13, has an interval that covers 0 in micro F1 (in macro F1, post hoc, +1.33 [+0.29, +2.36]): the pre-registered pass over the CRF head is a property that mention replacement already has on these units (post hoc). Three intervals in micro F1 comparing a names variant with MR3 lie above 0, none of them a decision: MaterioMiner coarse-grained (Table 3, 3 degrees of freedom), the 2-of-3-teachers ablation on 3 seeds (Supplementary Note 4) and SOFC-Exp (Results, *SOFC-Exp: no gain shown*). Per entity type the pattern holds for most types (Supplementary Note 6).

D7's protocol foresaw a gain of +1.5 to +1.65 from the seeds in hand, an 11-unit half-width of about 2 and a chance of a pass of about 0.4; "not established" is thus the likely outcome of a small effect under this design, not evidence of none. The interval excludes a gain above +2.26, nearly as large as mention replacement's whole gain (+2.46). The follow-up on MaterioMiner's strongest system gives a similar increment (Results, *The names on the strongest system*).

The names' source and the lexicon's settings

Source (D8). For each held-out subfield a mixed corpus of 40,000 sentences, drawn equally from the six other subfields' corpora and built the same way, holds 49–70% of the subfield's unseen test names against the topic corpus's 55–81%. MR6 + names with the topic's own corpus scores 51.73 against 51.91 with the mixed one, −0.18 [−2.01, +1.65] paired by subfield, in a comparison that detects 2.5 points. The mixed-corpus variant gains +3.38 [−0.89, +7.64] over the CRF head, +1.10 [−0.76, +2.97] over MR3 and +0.70 [−1.19, +2.58] over MR6 (pre-specified), its own lexicon holding 18.3% of the unseen test names. The protocol fixed this reading of a fail with the mixed variant above the CRF head: a large materials corpus of other topics does as well, a simpler method.

The condition holds on the point estimate (+3.38) but not on the interval, and "does as well" is an equivalence that a comparison detecting 2.5 points cannot establish. An amendment of the same protocol, made before any of D8's runs, had framed D8 more narrowly, as testing whether the names the topic adds, and the contexts they come in, are what the gain needs (the mixed corpora hold most of what the topic corpora hold); D8 answers that they are not detectably.

Settings (pre-specified, 3 seeds). Five ablations, of corpus size (10,000 and 20,000 sentences), threshold (0.7), agreement (2 of 3 teachers) and the names' share of the pool, keep micro F1 within 0.16 of MR6 + names on the same seeds (the screen's, on which it was selected, a slightly flattering reference), every interval against it within −2.45 and +2.14, while the lexicon ranges from 19 to 79 thousand distinct names, summed over the seven subfields, and from 16% to 32% of the unseen test names (Supplementary Note 4). Effects beyond those bounds are excluded; smaller ones are not.

What the names do

On MuLMS, pooled over the held-out subfields (post hoc, as is this section up to the recall-matched control; Table 4; Figure 3 puts each system's recall and precision on the unseen names on one plane with curves of equal F1), the CRF head finds 39.4% of the unseen names at 55.8% precision, MR3 43.1% at 54.3, MR6 43.7% at 53.9 and MR6 + names 47.3% at 50.9: against MR6, the copy-matched control, the names raise recall on unseen names by 3.6 points and lower precision by 3.0, a net +0.7 F1 on them, the point moving almost along a curve of equal F1. Paired by subfield on the decision seeds, the recall gain is +2.56 [+0.76, +4.36], the precision loss −2.22 [−5.03, +0.60] and the net +0.56 [−1.44, +2.56]: only the recall gain's interval lies above 0. On MaterioMiner's papers the pooled rates of both recall and precision on unseen names rise over MR6, within paired intervals that cover 0 (Supplementary Note 5).

Which names the gain falls on (post hoc; the analysis and its reading fixed before its run). The lexicons were not stored with the runs; an analysis of 31 October rebuilt each from its unit's own teachers and corpus, reproducing the stored teachers' scores and coverage counts in every unit, and split the unseen test names by whether the lexicon holds them (Table 5; Supplementary Note 5 gives the reading rule and the full analysis of 31 October (16:13)). On MuLMS the lexicon holds 22% of the unseen test names, pooled over the subfields (7% to 36% by subfield). On those, MR6 + names recalls 82.4% against MR6's 75.0%; on the others, 36.8% against 35.0%; the gain on the held names is the larger in every subfield, and the difference of the two gains has an interval above 0 (Table 5). The reading rule fixed in advance (Supplementary Note 5) calls this *name knowledge* if the other names' gain is read on its interval, which covers 0, and *mixed* if on its point estimate (+1.12); MaterioMiner's result is what the rule calls a *generic shift*, alike gains on both parts, on point estimates, and none of its outcomes on intervals. The copies alone show no detectable split (MR6 over the CRF head). Pooled precision falls on both parts (74.7% to 69.0% on the held names, 46.4% to 44.0% on the others), and two thirds of the wrong predictions the names add, and 45% of the correct ones, are names the lexicon does not hold. Because the teachers are the baseline, the lexicon holds the unseen names the baseline already finds most easily: the CRF head recalls 71.8% of them and 29.7% of the others (Table S3).

Against a recall-matched control (follow-up; D11, D12). A recall gain paid for in precision might be no more than a shift toward tagging more spans, which a constant bias at decoding would also give. The follow-up of 1 November tested this on MuLMS at the decision seeds (Supplementary Note 10). MR6 was run again and decoded with a bias D added to the score of every tag but O, over a grid from −2 to 5 (at D = 0 all 49 runs equal the stored ones in every score, in the training history and in the predictions); per subfield the bias D* was found at which MR6's recall over all test entities, pooled over the seeds, equals that of MR6 + names, every quantity of the control at D* interpolated linearly between the two grid values around it. At that bias the names find more of the unseen test names their lexicon holds, +11.24 [+3.31, +19.18], in every subfield (D11, passes; exact sign-flip P = 0.016, post hoc), and fewer of the other unseen names, −1.73 [−3.02, −0.44] (pre-specified; −0.88 [−2.63, +0.87] on the four matched subfields, a reading fixed in the analysis code of 11:04, before any result); in micro F1 they

are not shown to beat the control, +3.26 [−0.79, +7.31] (D12), and in precision +5.78 [−1.22, +12.79] (pre-specified) (Table 6, Figure 4). In three subfields, electrolysis, pemfc and sofc, no bias on the grid reaches the names' recall: MR6's recall peaks at D = 2.5 or 3, 0.2 to 0.5 points short, and falls beyond, where more spans are tagged but fewer match exactly (Table S8). The protocol matched these at the grid's end, D = 5, where MR6's recall is 1.7 to 2.9 points below the names', and its seed-mean F1 5.3 to 7.0 points and its recall on the held names 6.9 to 11.5 points below its own at D = 0, which favours the names in both decisions. The protocol also reported the decisions without these flagged subfields (pre-specified): D11 +6.05 [−3.33, +15.44], all four subfields positive with 3 degrees of freedom (an interval that could detect only about 12 points), and D12 −0.03 [−2.78, +2.72].

So +11.24 is the registered statistic, not the size of the effect. Per subfield it is the names' gain over MR6 at D = 0 on the held names, +7.29 on average (Table 5), plus MR6's own loss on them from D = 0 to D*, +3.96 [−1.06, +8.98], nearly all of it in the three flagged subfields (+0.14 on the other four; post hoc). Two post hoc readings that do not depend on the protocol's matching of those three agree in direction: with them matched at the grid value nearest the names' recall, D11 is +8.63 [+3.35, +13.90] and D12 +1.37 [−0.66, +3.41]; and against MR6 at whichever bias gives it its highest recall on the held names, the names lead by +6.52 [+2.15, +10.89], in every subfield (Figure 4), while MR6 at the bias best for the other unseen names overtakes them there, −4.16 [−5.82, −2.50] (Supplementary Note 10). No single decoding bias gives MR6 the names' recall on the names the lexicon holds. A lexicon is mostly materials names and values (83 to 99% of its distinct names; Supplementary Note 10), so a shift toward those types could also favour the held names. Within types, at D = 0, where no matching is involved, the names gain on the materials names the lexicon holds and not detectably on the others (+7.98 [+2.15, +13.82] against +0.89 [−3.48, +5.27]; post hoc; Table S9), and not detectably on the values it holds (+1.74 [−1.16, +4.65]), though values are most of every lexicon's distinct names; of the held names recalled beyond MR6, about two thirds are materials names and a quarter are of other types, whose held names are few but gain +9.80 [+3.34, +16.25]. The gain is thus consistent with the student having learned names from its lexicon, chiefly materials names (a post hoc reading, on MuLMS; which of those names its copies contained was not measured). At the protocol's matched bias it recalls fewer of the other unseen names, though not detectably on the four matched subfields, and it is not shown to gain F1 over the bias.

SOFC-Exp: no gain shown

On SOFC-Exp's five folds of held-out articles MR6 + names with the SOFC corpus scores 82.75 against the CRF head's 82.79, −0.04 [−1.21, +1.14] (D6). It gains +0.45 [+0.04, +0.86] over MR3 on every fold (pre-specified), but MR3 itself is not shown to gain there (−0.48 [−1.36, +0.39], a decision of 20 October (09:14); −0.29 [−1.04, +0.47] on the official split, paired by seed, a companion decision of 16 October (14:39), also failed). Two things differ on this corpus and cannot be separated here: only 46% of a fold's test names are unseen (81% fine- and 79% coarse-grained on MaterioMiner's papers, 88% on MuLMS's subfields) and the CRF head already finds 80% of them; and the baseline is at 82.8 F1 with a random-split gap of 3.5 points, so there is little headroom. The SOFC corpus holds fewer of the unseen test names (36%, 27–43% per fold, measured before the runs) than MuLMS's topic corpora (55–81%), but its lexicon holds 20–33% per fold, no less than MuLMS's or MaterioMiner's, and against MR3 the names' interval lies above 0 there (pre-specified, one of many comparisons without a decision): it is mention replacement that is not shown to gain. Whether training size explains that is open: at MaterioMiner's size MR3 gained +0.50 [−0.47, +1.46], also not shown to differ from 0 (a decision of 17 October, 07:28; Supplementary Note 7). Pooled over the 16 units of the three corpora the estimate of MR6 + names is +2.45 [+0.50, +4.40] and that of MR3 +1.46 [+0.28, +2.64], and over the 9 units of MaterioMiner and SOFC-Exp that of multi-task training with MR3 +2.93 [+0.14, +5.72] (pre-specified, MR3's and multi-task training's on 20 October at 09:14, the names' with D6 at 13:53); the claim stays the two-corpus one, SOFC-Exp a corpus where no gain is shown.

The names on the strongest system (follow-up)

The names had been tested only on the single-task CRF head. A follow-up protocol, committed on 31 October after D1 to D8 had been written up and before any of its runs, expecting a pass to be unlikely (Supplementary Note 1), tested them on MaterioMiner's strongest system, multi-task training with mention replacement: the names system against the multi-task model with six-copy mention replacement, over 10 seeds on the four held-out papers, fine-grained (D9) and coarse-grained (D10), its lexicon harvested as in the Methods (*Systems*) by three multi-task teachers with a filter over both heads (Supplementary Note 8). Neither decision passes (Table 2): fine-grained, all four papers are positive (0.16 to 1.62; two-sided t-test P = 0.052, and a sign-flip P of 0.125, the smallest four units allow); coarse-grained, three are. Over multi-task training with three-copy mention replacement the names' whole increment is +0.76 [−0.15, +1.68] and +0.91 [−0.14, +1.97] (pre-specified; Table S6): on this system too what the names add is about a point and not established. The fine-grained spread rests on 3 degrees of freedom, so the interval, not the 1.3 points it would detect at that spread (Table S7), is the result. The two-head filter's lexicons hold fewer of the unseen test names than the single-task lexicons of D3 and D4 (by the runs' stored counts, Supplementary Note 8: 7.5% against 9.6% fine-grained, 12.1% against 18.3% coarse-grained); whether that lowered the increment is not known.

What else was tried, and the count

MR6 + names was reached through two earlier protocols and a screen of seven variants on MuLMS's held-out subfields and their MaterioMiner analogues, each committed before its runs, with two rules applied to saved predictions (the lexicon as a dictionary, and label propagation within a document); Supplementary Note 7 gives them in full, with the count. In all, the protocols from 17 October (20:21) to 1 November took twenty decisions on the names or on mention replacement: the twelve of Table 2, the six of the two earlier protocols (four on MuLMS, two on MaterioMiner) and two on SOFC-Exp's folds. Five passed: D3, D4, D5, D11 (as registered, only with three subfields that no bias matched) and mention replacement's votes against the CRF head's votes. Of the seven decisions of Table 2 against a mention-replacement control or on the names' source, only D11, on recall, passes, as registered only with three subfields that no bias matched, its outcome chosen after the lexicon split had found the effect. If no method worked, about 0.5 of twenty would be expected to pass, whatever their dependence, which only makes chance passes come together, as the shared units of D3 to D5 do. The count leaves out the label-propagation rule's four decisions of 20 October, a rule on saved predictions rather than a use of names: with them it is twenty-four, six passing, about 0.6 expected by chance (Supplementary Note 7); the companion study's decisions, all earlier, are not counted.

Discussion

A tagger trained on a few annotated materials papers misses more than half of the entities of a new paper or subfield on MaterioMiner's fine-grained types and on MuLMS. Giving it names mined from unlabeled text by three agreeing teachers, through mention replacement, improves on the baseline on held-out documents of two corpora: on every MaterioMiner paper at both granularities (D3, D4) and pooled over 11 units (+3.58 [+0.95, +6.21]; D5). The pooled criterion was adopted mid-study, when mention replacement without names was already known to meet it, and MuLMS alone does not pass. Mention replacement without the names gains most of that on the same units (post hoc), and the names' own increment in F1, +0.80 [−0.66, +2.26] on the decision seeds (+0.83 [−0.01, +1.67] in macro and +0.89 [−0.40, +2.18] in weighted F1, post hoc), is not established, nor on MaterioMiner's strongest system. What the names do add, on MuLMS, is recall on the unseen names their lexicon holds: post hoc, the names model recalls more of them than MR6 does at any bias of the grid (+6.52 [+2.15, +10.89]). The registered recall-matched decision (D11) passes, but only with three subfields where no bias reached the names' overall recall (+6.05 [−3.33, +15.44] on the four where one did), and no F1 gain over the bias is shown (D12). This is consistent with the student having learned names from its lexicon, chiefly materials names; on MaterioMiner the reading rule fixed in advance reads a generic shift, on point estimates.

For practitioners who fine-tune an encoder tagger on a few hundred annotated sentences to read documents whose names are mostly unseen in training (no large language model was compared), the advice is mention replacement with three copies. Its gain on held-out documents passed pre-registered on MaterioMiner (companion study) and is above 0 over the 16 units of the three corpora (+1.46 [+0.28, +2.64], pre-specified) and over the 11 of MuLMS and MaterioMiner (+2.35 [+0.94, +3.75], post hoc), while on MuLMS alone only its votes passed (Supplementary Note 7). It costs about twice the baseline's training time and needs no corpus. Neither three more copies nor the harvested names are shown to add to it in micro F1, here or on MaterioMiner's strongest system; in the secondary measures, post hoc, the two together gain +1.33 [+0.29, +2.36] in macro and +1.25 [−0.29, +2.78] in weighted F1 over it on the decision seeds. A names student costs three to seven times an MR3 run, by how the harvest is charged, plus a 40,000-sentence corpus (Supplementary Note 2); if names are used at all, a corpus of other materials topics was not detectably worse than the held-out topic's own (D8). The names shift recall toward the unseen names their lexicon holds, by more than any decoding bias of the grid shifts MR6 (post hoc), at a cost on the other unseen names against a recall-matched bias (−1.73 [−3.02, −0.44], pre-specified; −0.88 [−2.63, +0.87] on the four subfields it matched) though not against MR6 as trained (+1.12 [−0.78, +3.02], post hoc), and they are not shown to gain F1 (Results, *What the names do*). They are worth their cost where those names matter more than F1: names the baseline's teachers already tag confidently, 72% of which the baseline itself finds (Table S3). Even the best system finds only half of a held-out MuLMS subfield's entities. If F1 is the goal, stop at mention replacement, and expect no gain to be shown on a corpus like SOFC-Exp, where half the test names are seen and the baseline is at 83 F1 (Results, *SOFC-Exp: no gain shown*).

Several limitations qualify these results. Seven subfields and four papers give 6 and 3 degrees of freedom; the subfields differ by 24 points under the CRF head (32.49 to 56.82) and range from 2 to 11 articles, the papers from 41 to 201 sentences, and the pooled decision weights them equally as if drawn from one population. The leave-one-out and sign-flip checks (Results, *The pre-registered decisions*) are post hoc, and with four units the sign-flip test cannot go below P = 0.125, so D3 and D4 rest on the t interval alone; a mixed model was not fitted, and with two corpora no between-corpus variance could be estimated, so the pooled intervals are conditional on these two corpora. The pooled criterion's condition that neither corpus's own interval lie below 0 is weak: at the observed spreads it would fail only if a corpus lost more than about 4.5 points on average (MuLMS) or 2.9 (MaterioMiner).

The criterion under which MR6 + names passes was adopted on 20 October on the author's decision to push for a methods claim, after part of the screen and before the names system's own screen result, and when mention replacement was already known to meet it (Methods, *Evaluation protocol*); under the earlier criterion there is no claim: D3, D4 and D6 would not have run, and D5 did not exist. The gate on D3 and D4 was last lifted at 18:42 that day, after the screen had chosen MR6 + names, by an amendment whose reading from the screen's spread put D5 at about [+0.1, +5.9]; the plan's estimate of a methods paper had fallen to about 0.2 that morning. The screen that chose MR6 + names and the MuLMS decisions share the same seven subfields: the decision seeds keep the selection's seed noise out of every MuLMS decision except D5 as executed, but not its fit to these units.

Each MuLMS model trains on a pool of 400 sentences (340 for training, 60 for validation) of two or three of the other subfields' articles (Table 1), a different set per held-out subfield with some articles recurring across sets; the 88% share of unseen names and the 24-point spread between subfields reflect that.

Every positive result is at about 340 training sentences; MuLMS's full pool was never used, SOFC-Exp's larger training set is where no gain is shown, and the literature predicts smaller gains for mention replacement as training grows. The claim is scoped to a few hundred annotated sentences.

The names' effect is only partly isolated. The lexicon-membership split (Results, *What the names do*) was run after D1 to D8, on rebuilt lexicons, and its reading rule does not say whether a gain is read on its point estimate or its interval. The recall-matched control (D11, D12) was registered after D1 to D10 were known, with the names' side already stored, and run on MuLMS only. Its grid could not reach the names' recall in three of the seven subfields, which the protocol matched at the grid's end, to the control's disadvantage; without them D11's interval covers 0, and the readings that match them otherwise are post hoc (Results, *What the names do*). It applies one bias to every

tag, so it separates name knowledge from a shift toward the lexicon's types only through the post hoc split within types, and past its peak (D = 2.5 or 3) the bias lengthens and merges spans; a bias on the tags that begin an entity was not tried. The name-knowledge pattern is MuLMS's: on MaterioMiner the split reads as a generic shift, and which lexicon names a student's copies contained was not measured. The lexicon's precision was not estimated (the precision of the predictions of lexicon names, 69 to 75% on MuLMS and 81 to 83% on MaterioMiner, is the nearest figure).

MR3's three copies and swap rate 0.5, MR6's six copies, the 0.9 threshold and the three-teacher agreement were fixed in advance, not tuned; Dai and Adel tune the copies (1 to 10) and the rate (0.1 to 0.7) on development data, and a tuned mention replacement could close or widen every gap in Table 3. The two-head filter of the follow-up of 31 October (19:15) admits fewer names than the single-task one, and a looser one was not tried.

Null results are not equivalences. The source decision and the ablations detect differences of about 2 to 3 points (Supplementary Note 9); smaller effects of the source or the settings are not excluded.

The gazetteer-feature and self-training variants were rejected on 3 seeds, and one encoder, MatSciBERT, was used. No large language model was compared as an extractor, a teacher or a generator of same-type names [34,35]; a generated pool is a different source from names already present in unlabeled text and needs its own matched test.

The corpora come from Europe PMC's open-access subset, found by title terms, with journal concentration and licence mixes given in Supplementary Note 3, and are rebuilt from hashed reports rather than redistributed. Pre-registration rests on the repository's dated commits, whose times the author's machine sets, not on an independent registry, though GitHub's own log of the pushes, archived in the repository, puts every protocol commit of Table S1 (15 October to 1 November) on GitHub 1 to 7 seconds after its commit time (Supplementary Note 1).

Several questions remain open. The first is a training-size curve on MuLMS, whose six other subfields hold about 8,000 labeled sentences that the 340-sentence regime leaves unused, since mention replacement's gains shrink with training size [29,31]. Two variants dismissed on MuLMS on 3 seeds deserve 10 there: self-training on the corpus's teacher-labeled sentences, the one variant of the screen whose interval against the CRF head cleared 0, though its MaterioMiner analogue, the companion study's self-training on the fatigue corpus over 10 seeds, is not shown to beat MR3 (−1.03 [−4.47, +2.41] fine-grained, +0.28 [−1.46, +2.01] coarse-grained, and its votes fall below MR3's coarse-grained, −1.32 [−1.85, −0.80]; post hoc, both sides' results known when it was named); and the lexicon as a gazetteer feature (Supplementary Note 7). A bias on the tags that begin an entity only, or one per entity type, might match the three subfields that one bias on every tag could not, since past its peak (D = 2.5 or 3) that bias lengthens and merges spans (Supplementary Note 10); decoding MR6 + names on the same grid would compare the two systems' whole recall–precision curves, and a lexicon stripped of the test names it holds would show whether the recall gain needs those very names. And the lexicon's precision, never estimated, could be read from a sample of 100 names per type judged by a materials scientist; it would tell whether the names fail to add F1 because they are wrong, redundant with the training mentions, or already within the baseline's reach.

Every decision, the criterion's revision and the amendments are reported. The practical advice is mention replacement. The methodological advice is to test a names method against a control with the same copies, and its recall against a decoding bias checked beforehand to reach the method's recall, on documents the training data has never seen, with enough seeds and units to detect a point or two.

Methods

Terms used throughout

- *Unit.* A held-out MuLMS subfield, MaterioMiner paper or SOFC-Exp fold of articles, on which a system trained without it is scored.

- *Seed.* The random seed of a training run: 10 per unit (offsets 0 to 900), or the 7 *decision seeds* (300 to 900) where a protocol used the other 3 for a *screen*, a 3-seed comparison that chooses among variants and decides nothing.
- *CRF head.* The baseline, MatSciBERT with a conditional random field output layer (Methods, *Systems*); *multi-task training* gives one encoder a CRF head for each of MaterioMiner's two type levels.
- *MR3, MR6.* Mention replacement with three or six copies of each training sentence, each mention swapped with probability 0.5 for one of the same type drawn from the *pool*: the training mentions, and for *MR6 + names* the lexicon too.
- *Teachers, marginal, lexicon, student.* Three CRF heads trained on the training sentences paired with a held-out unit (Table 1), never on the unit itself (seeds 0, 100 and 200), tag an unlabeled corpus; a tag's marginal is the probability a teacher gives it at its word; the lexicon is the mentions all three tag with the same span and type, every marginal at least 0.9; the student is the model trained on the gold sentences and their copies.
- *Decision, pre-specified, post hoc, follow-up.* A decision is a comparison whose verdict rule a protocol fixed before its results; a pre-specified comparison was fixed in a protocol without one; a post hoc one was specified after the results of the runs it analyses were known; the two *follow-ups* are the protocols of 31 October (19:15; D9, D10) and 1 November (10:51; D11, D12), registered after D1 to D8 had been written up. The *names route* is the protocols that led to and tested harvested names (17 October to 1 November).
- *Paired and pooled.* A paired difference is the mean over units of each unit's difference, each side the mean of its seeds (for a recall or precision, its rate over the seeds), with a 95% Student-t interval over the units. A pooled rate is a recall or precision from counts summed over seeds and units, without an interval; a pooled decision or estimate is one paired difference over the units of two or more corpora, with its interval.
- *Vote, sign-flip test.* A vote is the entity-level majority of the three models of a seed triple; the sign-flip test is the exact two-sided permutation test of a mean paired difference over all 2^n assignments of signs to the n units.
- *Decoding bias, recall-matched control.* A constant D added at decoding to the score of every tag but O (the tag of a word outside any entity), so that a trained model tags more spans (or, below 0, fewer); the recall-matched control is MR6 decoded with the bias at which its recall over all test entities equals that of MR6 + names or, where no bias on the grid of twelve values from −2 to 5 reaches it (a *flagged* subfield), at the grid's end (Results, *What the names do*).
- *Seen and unseen names* are defined under *Data* below.

Data

Table 1 gives the three corpora; the short names in parentheses are the ones the figures and tables use. All three corpora are released under open licences; our conversions and the held-out protocol are in the repository.

MaterioMiner's fine-grained level breaks ties among classes by token count, then name (its 27th and 28th classes tie at 24 entities); Supplementary Note 3 gives its papers' DOIs and MuLMS's kept entities by type.

Three details matter below. On MuLMS a flat tagger needs one level, so 31.4% of the entities are dropped with the conversion (14,632 of 46,584: nested in a kept entity, sharing its span or crossing it), and text extraction had damaged 276 tokens, which we repair; our MuLMS scores cannot be compared with the published nested ones. On MuLMS the training pool is capped at 400 sentences on purpose, so that MuLMS tests MaterioMiner's regime, a few hundred annotated sentences from two or three documents (SOFC-Exp, trained on 36 articles, does not; Results, *SOFC-Exp: no gain shown*); the other subfields hold about 8,000 labeled sentences that no run here uses (Discussion). SOFC-Exp's official test articles are tokenized differently from its training articles; the article folds used here cut every article with the training articles' tokenizer.

Unseen names. A *name* is the word string of an entity of any type, values such as "10 mA cm−2" included. A test entity is unseen when its words are annotated as no entity, of any type, in any training sentence, and seen when some training sentence annotates them with its type [18]; entities whose words are annotated only under another type, 0.0–2.1% of a corpus's test entities (up to 4.5% of one paper's), count as neither. A predicted entity is classed

by the same rule from its words and its predicted type, so a correct prediction has its gold entity's kind: recall on unseen names is the share of the unseen gold entities found, and precision the share of the predictions classed unseen that are correct. A name, gold or predicted, is in a lexicon when the lexicon holds its words with its type.

Evaluation protocol

Seeds and scoring. Every system compared in a decision has 10 seeds per held-out unit (offsets 0, 100, …, 900), except where a protocol reserved the screen's 3 seeds (0 to 200) for selection and decided on the other 7; the mixed-corpus variant of D8 and the recall-matched control of D11 and D12 have those 7 seeds, and the lexicon-setting ablations the screen's 3 (Results, *The names' source and the lexicon's settings*). The score is micro F1 at the entity level: an entity counts as found only with its exact span and type; precision is the share of predicted entities that are correct, recall the share of gold (annotated) entities found, F1 their harmonic mean, micro over all entities of a test set (seqeval [56] in its default mode, as the datasets' own code scores), in percentage points.

Decision rule. A method improves on its reference when the 95% Student-t interval of the mean paired micro-F1 difference lies above 0, the pairs being the held-out units (7 subfields, 4 papers, 5 folds), each side the mean of its seeds (D11 applies the rule to a recall; Results, *What the names do*); with 4 papers the interval has 3 degrees of freedom, with 7 subfields 6. A pooled decision takes the units of two corpora together (11 units, 10 degrees of freedom) and requires, in addition, that neither corpus's own interval lie below 0. Each decision is one-sided at 2.5%, so of the twelve decisions about 0.3 would pass if nothing worked (0.5 of the twenty counted at the end of the Results); the decisions are not independent (Results, *The pre-registered decisions*). At 80% power and the observed spreads the names against MR6 detect 2.0 points pooled over the 11 units and the names against the CRF head 3.5 to 3.7 (Supplementary Note 9 gives every decision and the main comparisons).

Pre-registration, and what changed. Each protocol was committed to the repository before its results, with its decisions and its pre-specified comparisons. All but three were committed before any run of the system under test had started: D5's (as executed), that of D7 and D8, and that of D11 and D12 (below). The references were stored runs whose results were known, except for D9 and D10, whose control ran in the same sessions. Amendments are marked with their reason and time, and Supplementary Note 11 lists every check, deviation and failure of the names route. Supplementary Note 1 gives the commits and the chronology, with GitHub's own log of the pushes, which puts every protocol commit on GitHub 1 to 7 seconds after its commit time. The protocols of the names route were fixed between 17 October and 1 November 2026. Three points must be disclosed.

First, the criterion for a methods claim was revised on 20 October at 09:14, on the author's decision, recorded in the protocol, to push for a methods paper. Under the original criterion MR6 + names has no claim, since D1 failed; D3, D4 and D6 would not have run, and D5 did not exist. The changes, in order (UTC; Supplementary Notes 1 and 11):

- *18 October, 04:52.* The plan makes the MaterioMiner decisions of the protocol of 18 October (01:13) the gate of the route: topic corpora only if one passes, otherwise the evaluation paper.
- *18 October, 11:00.* The protocol of 11:00 replaces that gate with a coverage check, saying those decisions cannot run before 24 October. A methods claim needs a pass against the CRF head on MuLMS alone (D1) and one on MaterioMiner; the MaterioMiner analogue (the same system on MaterioMiner's papers, with a corpus on fatigue; D3, D4) runs only if D1 passes.
- *20 October, 00:44.* The week-11 MaterioMiner decisions of 18 October (01:13), run after all, are recorded: both fail, so under the plan of 04:52 the route would have stopped. The screen is pushed at 02:35.
- *05:03.* The screen's first session, which had run the names variants on electrolysis and MR6 on every subfield, is pulled; the names variants' outputs are not recorded, nor is it recorded whether their scores were read. They are identical to their reruns in every score, the training history and the predictions, and there MR6 + names gained less than MR6 (+2.06 against +2.97 over the CRF head). A batch with all the names variants' jobs is pushed before the quota is checked; the user is asked to cancel it, and the plan of 05:06 moves the names

variants after 24 October. The batch is not cancelled; from 08:10 the plan expects Kaggle to stop it at the week's quota and to rerun it after 24 October.

- *08:11 and 08:13.* The screen's first part is recorded (MR6 and the two gazetteer-feature variants, none above MR3), and the plan's estimate of a methods paper falls from about 0.35–0.4 to about 0.2.
- *09:14.* The revision adds a pooled decision (D5) as a second route to the claim and lifts the gate on D3 and D4 and a stop rule (no fourth protocol on MuLMS). Mention replacement is known, from runs already made, to meet the pooled criterion, by a post hoc estimate: the protocol records +2.34 [+0.94, +3.75] with MaterioMiner's fine-grained papers (+2.35 in Table 3, from per-unit values rounded differently) and +1.80 [+0.39, +3.22] with the coarse-grained ones. The names of 18 October (01:13), from the other subfields' sentences on MuLMS and the fatigue corpus on MaterioMiner, meet it too (+2.39 [+0.71, +4.06], computed after the revision; post hoc; Supplementary Note 12). The names system's pooled result does not exist, and the screen has reported three of its seven variants, not MR6 + names.
- *13:35.* The screen's result is recorded, MR6 + names chosen; the batch of 05:03 had run on past the quota to its end at 12:25. Its +3.16 over the CRF head is read as clearing D1's bar of 2.4 to 2.9 points, and the plan puts D1 at about 0.5.
- *13:53.* An amendment re-imposes the gate on D3 and D4 and adds D6, SOFC-Exp's folds, on a pass of D1.
- *18:42.* An amendment lifts the gate on D3 and D4 again, on the ground that D5 needs the analogue whatever D1 gives, and widens D6's condition to a pass of D1 or D5. Its reading from the screen's own spread leaves D1 needing a mean gain near 5 and puts D5 at about [+0.1, +5.9]; the plan now puts D1 at about 0.3, D5 at about 0.4 and a methods paper at about 0.3.

D5's analysis was fixed before its data, but neither its adoption nor the last lifting of the gate on D3 and D4, whose papers D5 needs, was blind to related results. The conclusion that the names' own increment is not established (D2, D7, D9, D10) rests on neither.

Second, D5's protocol named MuLMS's decision seeds as the units' source and, in the same paragraph, "each unit its 10-seed mean". The analysis as executed (D5) averages in the screen's seeds 0 to 200, on which MR6 + names was selected and whose runs were under way at 09:14, their results not yet recorded; the decision-seed reading (D5′), which leaves them out, agrees (Table 2), and the decomposition in the Results (*Whose gain is it*) uses it.

Third, the protocol of D7 and D8 (29 October) was committed two minutes after its first batch was pushed and five hours before that batch's first result, but after MR6 + names' own results on the same units (D1, D3) were known: only MR6 and the mixed-corpus variant were new runs. That of D11 and D12 (1 November, 10:51), registered at the request of an internal review, run by the AI assistant, of the draft (*Use of an AI assistant*, below) after D1 to D10 and the lexicon split (Results, *What the names do*) were known, is of the same kind: the names' side is the stored runs, and only the control, MR6 run again (identically) and decoded under the bias, was new. That of D9 and D10 (31 October, 19:15) followed the write-up of D1 to D8 and preceded its runs. Every other comparison, here or from the earlier protocols, is labelled pre-specified or post hoc where it appears.

Runs. Kaggle T4 GPUs with deterministic CUDA, one frozen commit per batch, a pinned image and pinned libraries; model snapshots pinned by commit. From 15 October 16:00 UTC every GPU session that completed, but two, began by repeating an earlier run, and every repeat was identical (the sessions that failed or hung left no record of their jobs); the training settings are under *Systems*, and the compute and the repeats in Supplementary Note 2.

Systems

Figure 1 sketches MR6 + names, the last of the four systems compared throughout:

- **CRF head.** MatSciBERT [4] with a token classifier on the first sub-word of each word and a linear-chain conditional random field (CRF) over the words, constrained to the IOB2 tag grammar (a tag that begins an entity, continues it, or is outside one), trained by likelihood and decoded with Viterbi, the highest-scoring tag sequence [57,58], with early stopping on validation F1 (*Training*, below). It is the reference of D1 and D3 to

D6 and the baseline every system is measured from. It is not the strongest single model we know on every corpus: on MaterioMiner's held-out papers, fine-grained, multi-task training gains +3.72 [+0.57, +6.87] over it and multi-task training with three-copy mention replacement +6.73 [+6.03, +7.43] (pre-specified comparisons without a decision in the companion evaluation study; Table S1); neither transfers to SOFC-Exp, and MuLMS has one type level, so the CRF head is the reference the three corpora share. The follow-up of 31 October (19:15) tested the names on that system, with six copies (Results, *The names on the strongest system*).

- **MR3.** Mention replacement [29]: three copies of each training sentence with an entity, each mention replaced with probability 0.5 by a training mention of the same type; copies that change nothing are dropped. The student trains on the gold sentences and the copies with the CRF head's training settings and seed.
- **MR6.** The same with six copies: the matched control for the names.
- **MR6 + names.** MR6 whose replacement pool also holds a lexicon harvested from unlabeled text:
  - *Teachers.* The CRF head's own runs of seeds 0, 100 and 200 on the unit's training sentences, so the teachers are the baseline, not a stronger model.
  - *Corpus.* 40,000 sentences of open-access full texts from Europe PMC [59] found by a query on titles for the held-out subfield (for MaterioMiner, "fatigue" in the title and a metal or process term, such as alloy, steel, titanium, weld or selective laser melting, in the title or abstract; for MuLMS's solid-oxide subfield, "solid oxide", "solid-oxide", "SOFC", "SOEC" and their plurals; for SOFC-Exp, "solid oxide", "SOFC" and "SOFCs"; every query is in the corpus reports), with the datasets' own articles excluded by identifier (for MuLMS, which ships no identifiers, the 16 of its 50 articles found on Europe PMC; the rest are in journals Europe PMC does not index) and every sentence sharing a word 8-gram with a dataset sentence dropped (for MuLMS also a 50-character run). Sentences are split by a rule-based splitter that does not break after abbreviations or inside decimal numbers, and words by a regular expression that separates punctuation; sentences of 5 to 120 words with at least half alphabetic characters are kept, and formulas longer than 30 characters are cut. Each corpus has 290–368 articles with text (their composition in Supplementary Note 3). A topic corpus holds 55–81% of a held-out MuLMS subfield's unseen test names, the fatigue corpus 50–62% of a held-out paper's fine-grained (49–65% of its coarse-grained), the SOFC corpus 36% of the folds' (27–43% per fold; measured before the runs).
  - *Lexicon.* Each teacher tags the whole corpus and returns, per word, the probability the model assigns to the tag it chose (the forward–backward marginal of the Viterbi tag). A mention joins the lexicon when the three teachers tag the same span with the same type and every tag of the span has a marginal of at least 0.9 under each teacher, whatever the rest of the sentence. Per held-out MuLMS subfield this admits 8,328–50,869 mentions (4,380–13,395 distinct names), which hold 22.2% of the unseen test names pooled over the subfields (7–36% per subfield), about a third of what the corpus holds. The lexicon's precision was not estimated and its names were not stored with the runs (Discussion).
  - *Pool.* The training mentions (601–1,309 per MuLMS unit) and the harvested mentions, one entry per occurrence in the corpus, so that a draw is uniform over entries and a frequent name is drawn more often; harvested entries are at least 92% of a unit's pool. A mention is replaced only by an entry of its own type, and the lexicons are chiefly materials names and values (83 to 99% of their distinct names; Supplementary Note 10), so mentions of other types are replaced mostly by training mentions; the share of replacements drawn from the lexicon was not measured.

An example. Take a training sentence, constructed for illustration, "The Pt catalyst lowers the overpotential", with Pt annotated MAT and overpotential PROPERTY. For the held-out electrolysis subfield, a copy made by MR6 might read "The Ni catalyst lowers the overpotential", the replacement drawn from the unit's training mentions, and a copy made by MR6 + names "The MoS2 catalyst lowers the Tafel slope", both replacements drawn from the subfield's lexicon. Its most frequent entries are, as MAT, Ni, Pt, hydrogen, water, electrode, Co and catalyst, and, as PROPERTY, current density and scan rate (each also in the plural) and Tafel slope, and, as VALUE, 10 mA cm−2, 30 min, 2 h, 60 °C and 1 M; values are 70% of its distinct names and 30% of its entries, materials 23% and

64% (the rebuilt lexicon, which is not released; Results, *What the names do*; Supplementary Note 10); a materials scientist might not count electrode or catalyst as materials, and the lexicon's precision was not estimated.

Every run stores its teachers' scores, the lexicon's counts and the share of the unseen test names it holds, the corpus's hash and the filter. A names student costs about 6 times the CRF head's run with the harvest shared over its job's three or four students and about 14 times charged to one student, not counting the corpus's construction (Supplementary Note 2).

Training. Every system trains MatSciBERT (110 million parameters) with the CRF head's settings: AdamW [60] with a learning rate of 5×10^{-5} and weight decay 0.01 for the encoder and 5×10^{-3} without weight decay for the CRF layer, a linear schedule that warms up over the first 10% of the steps of 20 epochs and then decays to 0, batch size 16, dropout 0.1, the gradient norm clipped at 1.0 and inputs truncated at 512 sub-words. Training stops when validation micro F1 has not improved for 3 epochs (at most 20), and the best epoch's weights are evaluated. The teachers' marginals come from the forward–backward algorithm.

The recall-matched control (D11, D12). MR6's stored decision jobs on MuLMS (400 pool sentences, six copies, seeds 300 to 900) were run again, and each run was decoded at every bias D of the grid −2, −1, −0.5, 0, 0.5, 1, 1.5, 2, 2.5, 3, 4 and 5, D added to the emission score of every tag but O before Viterbi decoding, training untouched. Per subfield, recall over all test entities is pooled over the seven seeds (correct entities summed over the seeds, divided by gold entities summed), for MR6 + names and for the control at each D. The matched bias D* is the first crossing of the names' recall by the control's, going from D = 0 toward it along the grid, found by linear interpolation between the two grid values that bracket it, and every quantity of the control at D* is the same interpolation of its grid values (pooled counts for the pooled rates, each run's micro F1 for the seed mean). A subfield whose target lies outside the control's range over the grid is matched at the nearer end of the grid and flagged; it stays in the decisions, which are also reported without the flagged subfields. D11 is recall on the unseen test names the lexicon holds and D12 micro F1, each the names minus the control at D*, paired over the seven subfields. D* is chosen on the test set by design: the control tests a mechanism and is not a calibration one could deploy (a bias chosen on validation folds, in the companion study, gained nothing; Supplementary Note 5).

Statistics and reproducibility

Tests. The decision rule (*Evaluation protocol*) is a one-sample Student-t test of the mean paired difference against 0: its interval lies above 0 exactly when the one-sided P value is below 0.025, that is, when the two-sided P value is below 0.05 with a positive mean. A test's n is its number of held-out units (4 papers, 5 folds, 7 subfields or 11 pooled units; Table 2), each unit's value the difference of the two systems' means over the unit's 10 seeds (7 on MuLMS's decision seeds), so seeds are not counted as independent observations. Table 2 gives each decision's two-sided P value, and Table S16 its t statistic, degrees of freedom (n − 1; Supplementary Note 13), one- and two-sided P values and the standard deviation of its per-unit differences, with the exact sign-flip P value beside them. They were computed on 1 November for the journal's reporting rules, from the per-unit differences the decisions used, and change no decision.

Assumptions. The t test assumes roughly normal per-unit differences. With 4 to 11 units a test of normality has little power, and none was run. The exact two-sided sign-flip test, which assumes only that each difference is symmetric about 0 under the null hypothesis, is given beside it (post hoc): it agrees for the pooled decisions (P = 0.011 for D5, 0.007 for D5′) and for D11 (0.016), but with 4 units it cannot go below 0.125, so D3 and D4 rest on the t test's assumption. Figure 2 and Supplementary Note 12 show the per-unit values; the Discussion gives what the pooled intervals assume about the corpora.

Multiplicity. No level was adjusted for the number of decisions: the protocols registered each rule unadjusted, and the count is reported instead (Results, *What else was tried, and the count*). Under a Bonferroni bound of 0.05/12 none of the four passes would hold (Results, *The pre-registered decisions*).

Estimates. Every other interval in the paper is the 95% Student-t interval of a paired difference over held-out units (*Terms used throughout*). Such intervals accompany reported and post hoc comparisons as estimates, not tests; "clears 0" says only that the lower bound is above 0. Pooled rates carry no interval. Supplementary Note 9 gives the difference each decision could detect with 80% power.

Reproducibility. Every run is deterministic, and every repeat of a run was identical (*Evaluation protocol*; Supplementary Note 2). The repository holds the code, every results file with its seeds and saved predictions, and the scripts behind every number (Code availability).

Use of an AI assistant

An AI assistant was used, under the author's direction, to draft the protocols, write the code, run the experiments, draft the text of every section from the results files and run adversarial internal reviews of the drafts, among them the review that asked for the recall-matched control (D11, D12). Every result in the paper was taken from the outputs of the repository's analysis scripts, and every date, odds figure and protocol-quoted number from its dated records, and checked against them by those reviews, which the same assistant ran. The author took the study's decisions, among them the revision of the criterion on 20 October and the approval of the recall-matched control's runs on 1 November, and is responsible for the content.

Relation to a companion study

This paper shares data, baselines and protocol with a companion evaluation study of the same programme (H.M., in preparation; its draft is in the repository), which reports the random-split gap, seed variance and the other additions tested. The two papers overlap in the Methods. The results that appear in both are the random-split gap and the unseen-name shares of the Introduction, mention replacement's MaterioMiner decisions and the multi-task results in the Methods (*Systems*), mention replacement on SOFC-Exp's official split and at MaterioMiner's training size (Results, *SOFC-Exp: no gain shown*), the recall calibration of Supplementary Note 5, the label-propagation rule of Supplementary Note 7, which the companion reports in its survey, and self-training on the fatigue corpus over MaterioMiner's papers (the companion's decision of 15 October), whose comparison with MR3 the Discussion quotes; the companion summarises D1–D8 and the decomposition in the Results (*Whose gain is it*) in one table and cites this paper for them.

Data availability

The work uses published corpora under their licences and open-access full texts for the unlabeled corpora, which are not redistributed. No personal data is involved. The same repository holds every results file with its seeds, environment and saved test predictions, and the corpus reports with their queries, article identifiers, licences and SHA-256 hashes. MaterioMiner's files are included under CC BY 4.0; SOFC-Exp is converted from the authors' repository (CC BY 4.0); MuLMS is read from its Hugging Face release at a pinned revision (CC BY-SA 4.0). The unlabeled corpora are not redistributed, 10–33% of their articles being under non-commercial or no-derivative licences; they rebuild from their reports, with hash checks, and the reports identify the CC BY articles. The runs did not store the harvested lexicons; `ner_baseline.py --students 0 --save-names` rebuilds each from its corpus (the lexicon-split analysis of 31 October), and they are not released, because they derive from corpora that are not redistributed.

Code availability

The repository, github.com/hatemadi15/AI_Research_Ontologies_GNN_GRL (Apache 2.0; the commit this paper describes will be tagged, and an archived copy with a DOI deposited, at submission), holds the code, every protocol with its dates and amendments, the analyses behind every number, the generated tables and figures, and the Kaggle batch definitions that produced each results file.

Acknowledgements

The author thanks the creators of MaterioMiner, SOFC-Exp and MuLMS for releasing their corpora under open licences. This study received no funding; the GPU compute was Kaggle's free tier.

Author contributions

H.M. conceived and directed the study and is responsible for its content; an AI assistant, under H.M.'s direction, drafted the protocols, wrote the code, ran the experiments and drafted the paper (Methods, *Use of an AI assistant*).

Competing interests

The author declares no financial or non-financial competing interests.

References

1. Durmaz, A. R., Thomas, A., Mishra, L., Murthy, R. N. & Straub, T. An ontology-based text mining dataset for extraction of process-structure-property entities. *Scientific Data* **11**, 1112 (2024). https://doi.org/10.1038/s41597-024-03926-5
2. Friedrich, A. et al. The SOFC-Exp Corpus and Neural Approaches to Information Extraction in the Materials Science Domain. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* 1255–1268 (Association for Computational Linguistics, 2020). https://doi.org/10.18653/v1/2020.acl-main.116
3. Schrader, T. P., Finco, M., Grünewald, S., Hildebrand, F. & Friedrich, A. MuLMS: A Multi-Layer Annotated Text Corpus for Information Extraction in the Materials Science Domain. In *Proceedings of the Second Workshop on Information Extraction from Scientific Publications* 84–100 (Association for Computational Linguistics, 2023). https://doi.org/10.18653/v1/2023.wiesp-1.11
4. Gupta, T., Zaki, M., Krishnan, N. M. A. & Mausam. MatSciBERT: A materials domain language model for text mining and information extraction. *npj Computational Materials* **8**, 102 (2022). https://doi.org/10.1038/s41524-022-00784-w
5. Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* 4171–4186 (Association for Computational Linguistics, 2019). https://doi.org/10.18653/v1/N19-1423
6. Olivetti, E. A. et al. Data-driven materials research enabled by natural language processing and information extraction. *Applied Physics Reviews* **7**, 041317 (2020). https://doi.org/10.1063/5.0021106
7. Kononova, O. et al. Opportunities and challenges of text mining in materials research. *iScience* **24**, 102155 (2021). https://doi.org/10.1016/j.isci.2021.102155
8. Swain, M. C. & Cole, J. M. ChemDataExtractor: A Toolkit for Automated Extraction of Chemical Information from the Scientific Literature. *Journal of Chemical Information and Modeling* **56**, 1894–1904 (2016). https://doi.org/10.1021/acs.jcim.6b00207
9. Weston, L. et al. Named Entity Recognition and Normalization Applied to Large-Scale Information Extraction from the Materials Science Literature. *Journal of Chemical Information and Modeling* **59**, 3692–3702 (2019). https://doi.org/10.1021/acs.jcim.9b00470
10. Trewartha, A. et al. Quantifying the advantage of domain-specific pre-training on named entity recognition tasks in materials science. *Patterns* **3**, 100488 (2022). https://doi.org/10.1016/j.patter.2022.100488
11. Song, Y., Miret, S. & Liu, B. MatSci-NLP: Evaluating Scientific Language Models on Materials Science Language Tasks Using Text-to-Schema Modeling. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* 3621–3639 (Association for Computational Linguistics, 2023). https://doi.org/10.18653/v1/2023.acl-long.201

12. Dagdelen, J. et al. Structured information extraction from scientific text with large language models. *Nature Communications* **15**, 1418 (2024). https://doi.org/10.1038/s41467-024-45563-x
13. Polak, M. P. & Morgan, D. Extracting accurate materials data from research papers with conversational language models and prompt engineering. *Nature Communications* **15**, 1569 (2024). https://doi.org/10.1038/s41467-024-45914-8
14. Potu, S. T. et al. Ontology-conformal recognition of materials entities using language models. *Scientific Reports* **15**, 18597 (2025). https://doi.org/10.1038/s41598-025-03619-y
15. Babl, F., Hennen, M., Murauer, J. & Geierhos, M. Random Splitting Negatively Impacts NER Evaluation: Quantifying and Eliminating the Overestimation of NER Performance. In *Findings of the Association for Computational Linguistics: ACL 2025* 9724–9738 (Association for Computational Linguistics, 2025). https://doi.org/10.18653/v1/2025.findings-acl.504
16. Gorman, K. & Bedrick, S. We Need to Talk about Standard Splits. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* 2786–2791 (Association for Computational Linguistics, 2019). https://doi.org/10.18653/v1/P19-1267
17. Søgaard, A., Ebert, S., Bastings, J. & Filippova, K. We Need To Talk About Random Splits. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume* 1823–1832 (Association for Computational Linguistics, 2021). https://doi.org/10.18653/v1/2021.eacl-main.156
18. Augenstein, I., Derczynski, L. & Bontcheva, K. Generalisation in named entity recognition: A quantitative analysis. *Computer Speech & Language* **44**, 61–83 (2017). https://doi.org/10.1016/j.csl.2017.01.012
19. Fu, J., Liu, P., Zhang, Q. & Huang, X. Rethinking Generalization of Neural Models: A Named Entity Recognition Case Study. In *Proceedings of the AAAI Conference on Artificial Intelligence* 7732–7739 (2020). https://doi.org/10.1609/aaai.v34i05.6276
20. Kim, H. & Kang, J. How Do Your Biomedical Named Entity Recognition Models Generalize to Novel Entities?. *IEEE Access* **10** (2022). https://doi.org/10.1109/ACCESS.2022.3157854
21. Lin, H. et al. A Rigorous Study on Named Entity Recognition: Can Fine-tuning Pretrained Model Lead to the Promised Land?. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* 7291–7300 (Association for Computational Linguistics, 2020). https://doi.org/10.18653/v1/2020.emnlp-main.592
22. Ghaddar, A., Langlais, P., Rashid, A. & Rezagholizadeh, M. Context-aware Adversarial Training for Name Regularity Bias in Named Entity Recognition. *Transactions of the Association for Computational Linguistics* **9**, 586–604 (2021). https://doi.org/10.1162/tacl_a_00386
23. Taillé, B., Guigue, V. & Gallinari, P. Contextualized Embeddings in Named-Entity Recognition: An Empirical Study on Generalization. In *Advances in Information Retrieval: 42nd European Conference on IR Research (ECIR 2020), Part II* 383–391 (Springer, 2020). https://doi.org/10.1007/978-3-030-45442-5_48
24. Agarwal, O., Yang, Y., Wallace, B. C. & Nenkova, A. Entity-Switched Datasets: An Approach to Auditing the In-Domain Robustness of Named Entity Recognition Models. (2020). https://arxiv.org/abs/2004.04123
25. Reimers, N. & Gurevych, I. Reporting Score Distributions Makes a Difference: Performance Study of LSTM-networks for Sequence Tagging. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing* 338–348 (Association for Computational Linguistics, 2017). https://doi.org/10.18653/v1/D17-1035
26. Dror, R., Baumer, G., Shlomov, S. & Reichart, R. The Hitchhiker's Guide to Testing Statistical Significance in Natural Language Processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* 1383–1392 (Association for Computational Linguistics, 2018). https://doi.org/10.18653/v1/P18-1128
27. Bouthillier, X. et al. Accounting for Variance in Machine Learning Benchmarks. In *Proceedings of Machine Learning and Systems* 747–769 (2021). Preprint at arXiv:2103.03098.
28. Card, D. et al. With Little Power Comes Great Responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* 9263–9274 (Association for Computational

Linguistics, 2020). https://doi.org/10.18653/v1/2020.emnlp-main.745
29. Dai, X. & Adel, H. An Analysis of Simple Data Augmentation for Named Entity Recognition. In *Proceedings of the 28th International Conference on Computational Linguistics* 3861–3867 (International Committee on Computational Linguistics, 2020). https://doi.org/10.18653/v1/2020.coling-main.343
30. Mysore, S. et al. The Materials Science Procedural Text Corpus: Annotating Materials Synthesis Procedures with Shallow Semantic Structures. In *Proceedings of the 13th Linguistic Annotation Workshop* 56–64 (Association for Computational Linguistics, 2019). https://doi.org/10.18653/v1/W19-4007
31. Torres, A. E., de Moura, E. S., da Silva, A. S., Nascimento, M. A. & Mesquita, F. An Experimental Study on Data Augmentation Techniques for Named Entity Recognition on Low-Resource Domains. (2024). Preprint at arXiv:2411.14551. https://arxiv.org/abs/2411.14551
32. Sharma, S. et al. When and how to paraphrase for named entity recognition?. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* 7052–7087 (Association for Computational Linguistics, 2023). https://doi.org/10.18653/v1/2023.acl-long.390
33. Bartolini, I. et al. Named Entity Recognition using context similarity data augmentation. In *CEUR Workshop Proceedings* (2024).
34. Zhou, R. et al. MELM: Data Augmentation with Masked Entity Language Modeling for Low-Resource NER. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* 2251–2262 (Association for Computational Linguistics, 2022). https://doi.org/10.18653/v1/2022.acl-long.160
35. Hu, X. et al. Entity-to-Text based Data Augmentation for various Named Entity Recognition Tasks. In *Findings of the Association for Computational Linguistics: ACL 2023* (Association for Computational Linguistics, 2023).
36. Cai, J. et al. Improving Low-resource Named Entity Recognition with Graph Propagated Data Augmentation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)* (Association for Computational Linguistics, 2023).
37. Amalvy, A., Labatut, V. & Dufour, R. Remplacement de mentions pour l'adaptation d'un corpus de reconnaissance d'entités nommées à un domaine cible. In *Actes de la 29e Conférence sur le Traitement Automatique des Langues Naturelles. Volume 1 : conférence principale* 198–205 (2022).
38. Ding, B. et al. DAGA: Data Augmentation with a Generation Approach for Low-resource Tagging Tasks. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* 6045–6057 (Association for Computational Linguistics, 2020). https://doi.org/10.18653/v1/2020.emnlp-main.488
39. Chen, J., Wang, Z., Tian, R., Yang, Z. & Yang, D. Local Additivity Based Data Augmentation for Semi-supervised NER. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* 1241–1251 (Association for Computational Linguistics, 2020). https://doi.org/10.18653/v1/2020.emnlp-main.95
40. Chen, S., Aguilar, G., Neves, L. & Solorio, T. Data Augmentation for Cross-Domain Named Entity Recognition. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, 2021).
41. Liu, Z. et al. CrossNER: Evaluating Cross-Domain Named Entity Recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence* 13452–13460 (2021).
42. Riloff, E. & Jones, R. Learning Dictionaries for Information Extraction by Multi-Level Bootstrapping. In *Proceedings of the Sixteenth National Conference on Artificial Intelligence (AAAI-99)* 474–479 (AAAI Press, 1999).
43. Kozareva, Z. Bootstrapping Named Entity Recognition with Automatically Generated Gazetteer Lists. In *Proceedings of the Eleventh Conference of the European Chapter of the Association for Computational Linguistics: Student Research Workshop* 15–21 (Association for Computational Linguistics, 2006).
44. Blum, A. & Mitchell, T. Combining labeled and unlabeled data with co-training. In *Proceedings of the Eleventh Annual Conference on Computational Learning Theory* 92–100 (1998). https://doi.org/10.1145/279943.279962

45. Zhou, Z.-H. & Li, M. Tri-training: exploiting unlabeled data using three classifiers. *IEEE Transactions on Knowledge and Data Engineering* **17**, 1529–1541 (2005). https://doi.org/10.1109/TKDE.2005.186
46. Zafarian, A. & Asghari, H. Improving NER Models by Exploiting Named Entity Gazetteer as External Knowledge. In *Proceedings of the First International Workshop on NLP Solutions for Under Resourced Languages (NSURL 2019)* (2019).
47. Meng, T., Fang, A., Rokhlenko, O. & Malmasi, S. GEMNET: Effective Gated Gazetteer Representations for Recognizing Complex Entities in Low-context Input. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* 1499–1512 (Association for Computational Linguistics, 2021). https://doi.org/10.18653/v1/2021.naacl-main.118
48. Song, C. H., Lawrie, D., Finin, T. & Mayfield, J. Improving Neural Named Entity Recognition with Gazetteers. In *Proceedings of the Thirty-Third International Florida Artificial Intelligence Research Society Conference (FLAIRS-33)* (AAAI Press, 2020).
49. Shang, J. et al. Learning Named Entity Tagger using Domain-Specific Dictionary. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing* 2054–2064 (Association for Computational Linguistics, 2018). https://doi.org/10.18653/v1/D18-1230
50. Liang, C. et al. BOND: BERT-Assisted Open-Domain Named Entity Recognition with Distant Supervision. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '20)* 1054–1064 (ACM, 2020). https://doi.org/10.1145/3394486.3403149
51. Lison, P., Barnes, J., Hubin, A. & Touileb, S. Named Entity Recognition without Labelled Data: A Weak Supervision Approach. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* 1518–1533 (2020). https://doi.org/10.18653/v1/2020.acl-main.139
52. Wu, Q., Lin, Z., Karlsson, B. F., Lou, J.-G. & Huang, B. Single-/Multi-Source Cross-Lingual NER via Teacher-Student Learning on Unlabeled Data in Target Language. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* 6505–6514 (Association for Computational Linguistics, 2020). https://doi.org/10.18653/v1/2020.acl-main.581
53. Qu, X. et al. Distantly-Supervised Named Entity Recognition with Adaptive Teacher Learning and Fine-grained Student Ensemble. In *Proceedings of the AAAI Conference on Artificial Intelligence* (2023).
54. Du, J. et al. Self-training Improves Pre-training for Natural Language Understanding. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* 5408–5418 (Association for Computational Linguistics, 2021). https://doi.org/10.18653/v1/2021.naacl-main.426
55. van Miltenburg, E., van der Lee, C. & Krahmer, E. Preregistering NLP research. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* 613–623 (Association for Computational Linguistics, 2021). https://doi.org/10.18653/v1/2021.naacl-main.51
56. Nakayama, H. seqeval: A Python framework for sequence labeling evaluation. (2018). https://github.com/chakki-works/seqeval
57. Lafferty, J., McCallum, A. & Pereira, F. Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data. In *Proceedings of the 18th International Conference on Machine Learning (ICML 2001)* 282–289 (2001).
58. Lample, G., Ballesteros, M., Subramanian, S., Kawakami, K. & Dyer, C. Neural Architectures for Named Entity Recognition. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* 260–270 (Association for Computational Linguistics, 2016). https://doi.org/10.18653/v1/N16-1030
59. The Europe PMC Consortium. Europe PMC: a full-text literature database for the life sciences and platform for innovation. *Nucleic Acids Research* **43**, D1042–D1048 (2015). https://doi.org/10.1093/nar/gku1061
60. Loshchilov, I. & Hutter, F. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations* (2019). Preprint at arXiv:1711.05101.

Figure legends

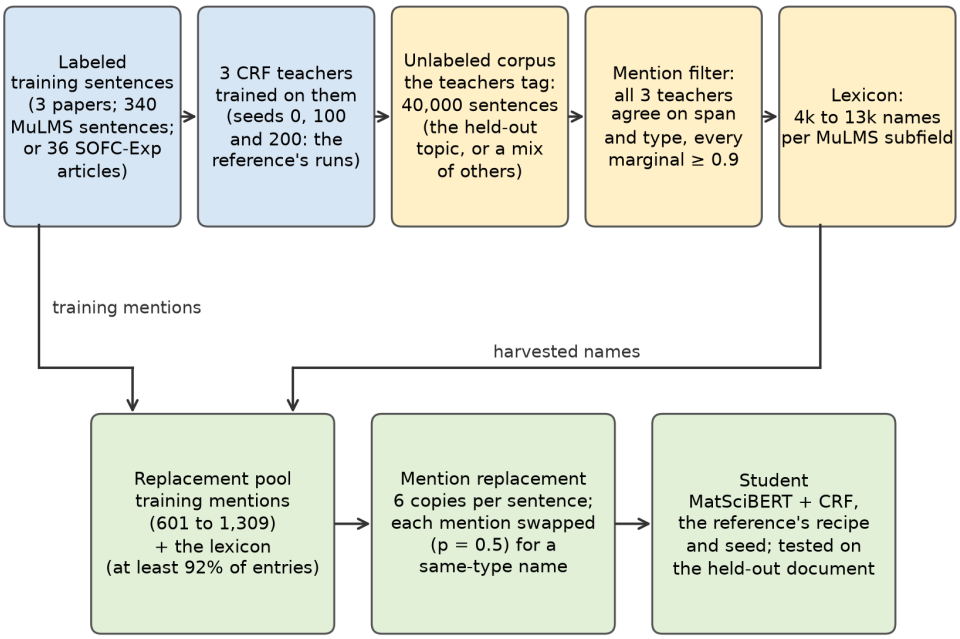


Figure 1. The names method, MR6 + names. Three CRF-head teachers trained on the labeled sentences tag a 40,000-sentence corpus; the mentions all three tag alike with every marginal at 0.9 or more form the lexicon; the replacement pool holds the training mentions and the lexicon; mention replacement writes six copies of each training sentence with its mentions swapped; the student trains on the gold sentences and the copies.

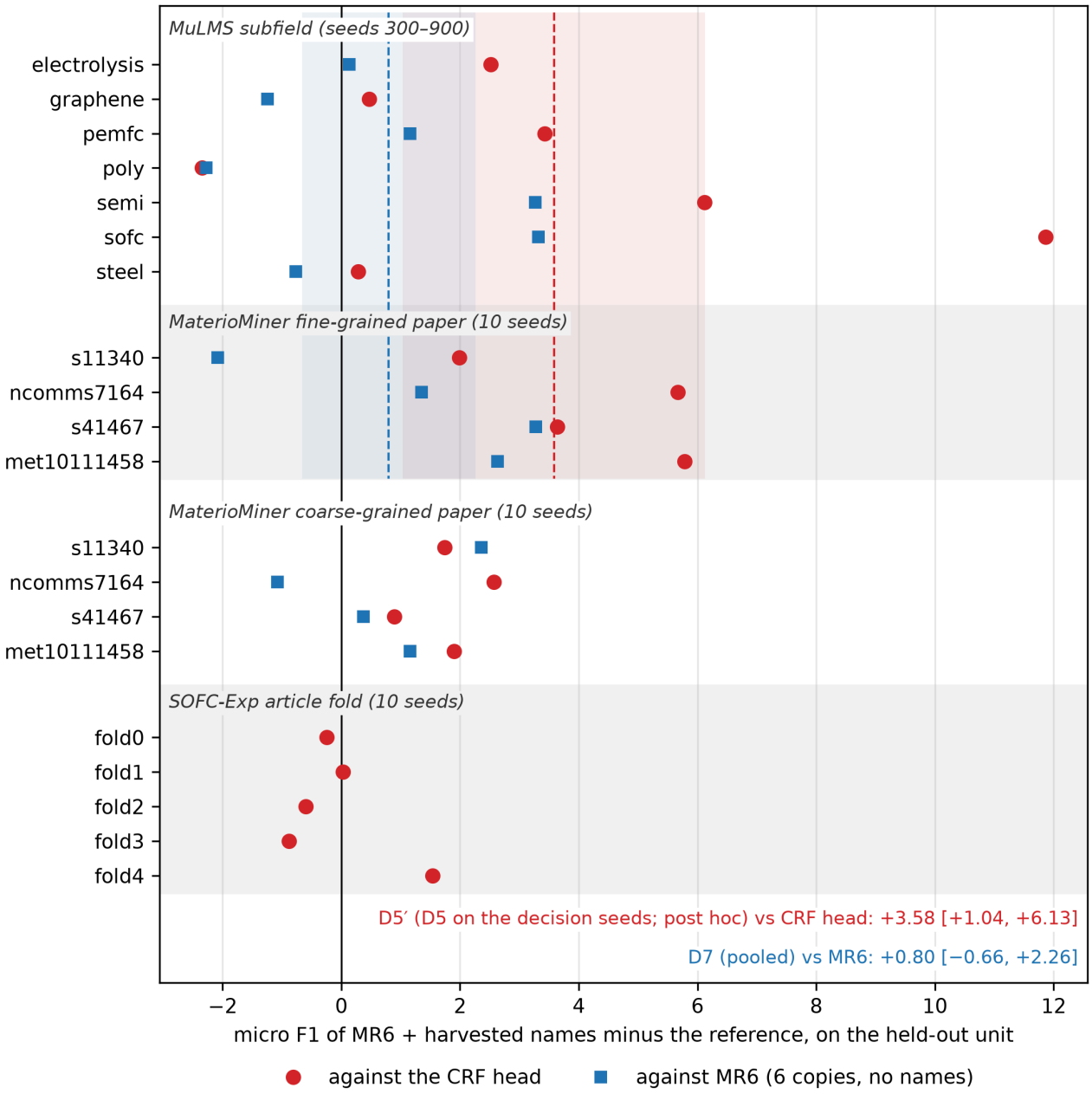


Figure 2. The gain of MR6 + names on each held-out unit. The micro-F1 difference of MR6 + names from the CRF head (red circles) and from MR6 (blue squares), each the difference of the two systems' seed means on the unit: MuLMS's 7 subfields on the decision seeds, MaterioMiner's 4 papers at both granularities and SOFC-Exp's 5 folds, where MR6 was not run (the protocol put it at about 13 hours of one T4 GPU, on a corpus where the names method does not pass; Supplementary Note 11). The shaded bands, drawn across the 11 units' rows, are the 95% Student-t intervals of the pooled differences over those units, D5′ (red; the post hoc decision-seed reading of D5) and D7 (blue), with dashed lines at their means; the units carry no error bars.

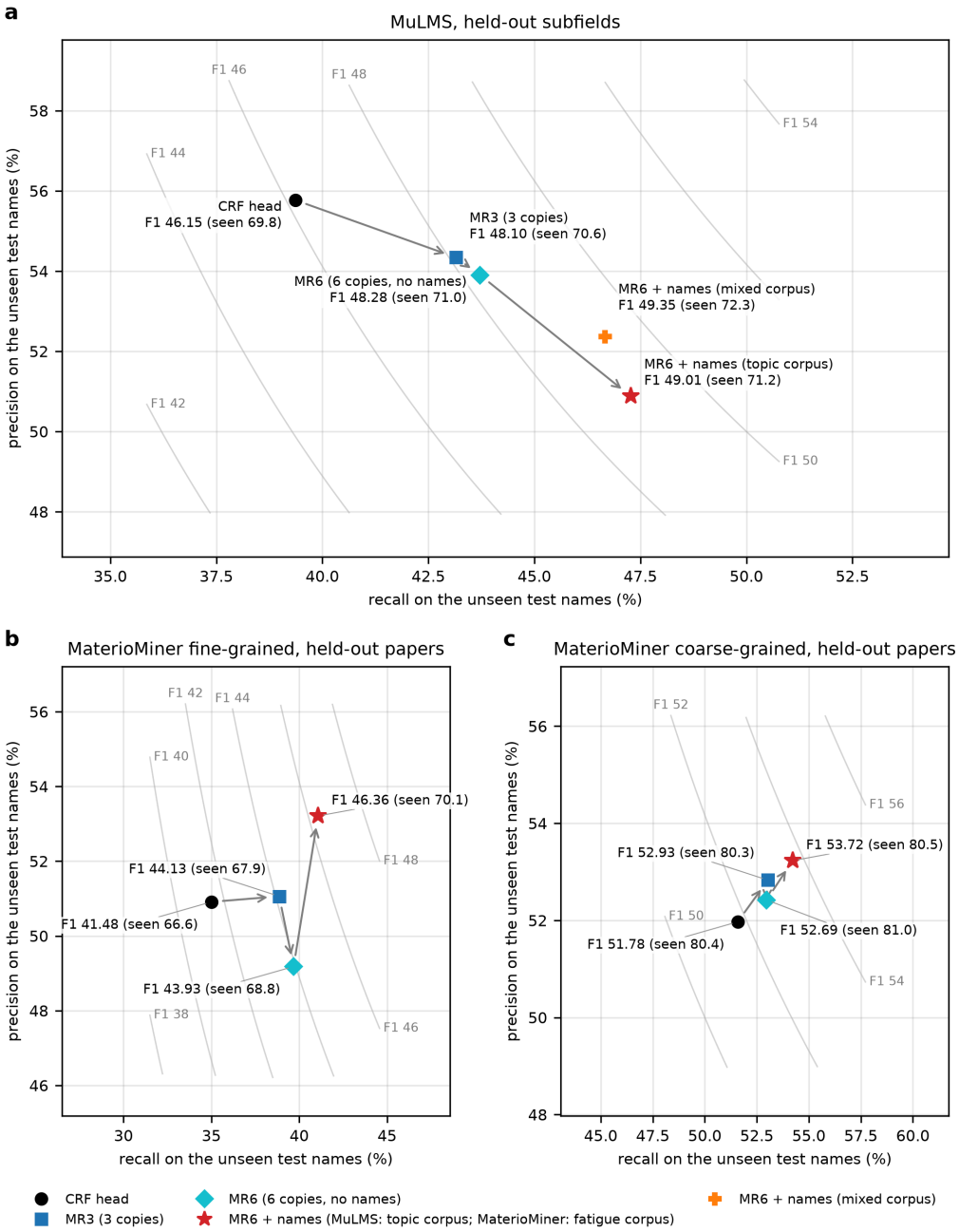


Figure 3. Recall and precision on the unseen test names. Each point pools the predictions over seeds and held-out units: the CRF head (black circles), MR3 (blue squares), MR6 (cyan diamonds) and MR6 + names (red stars) on **a**, MuLMS's held-out subfields, with the topic corpus and, orange plus sign, a mixed corpus; **b**, MaterioMiner's held-out papers, fine-grained; and **c**, the same papers, coarse-grained, with the fatigue corpus. Grey lines are iso-F1 curves; a grey arrow joins each system to the one it builds on (the CRF head, MR3, MR6, then the names). A point's label gives its F1 on the unseen names and, in parentheses, on the seen ones; in **b** and **c** the labels carry the scores only. Pooled rates carry no error bars. The mixed-corpus point uses seeds 300–900 (49 runs) and has no arrow, since MR6's point uses all ten seeds; on the same seeds the two compare as in Table 4's note.

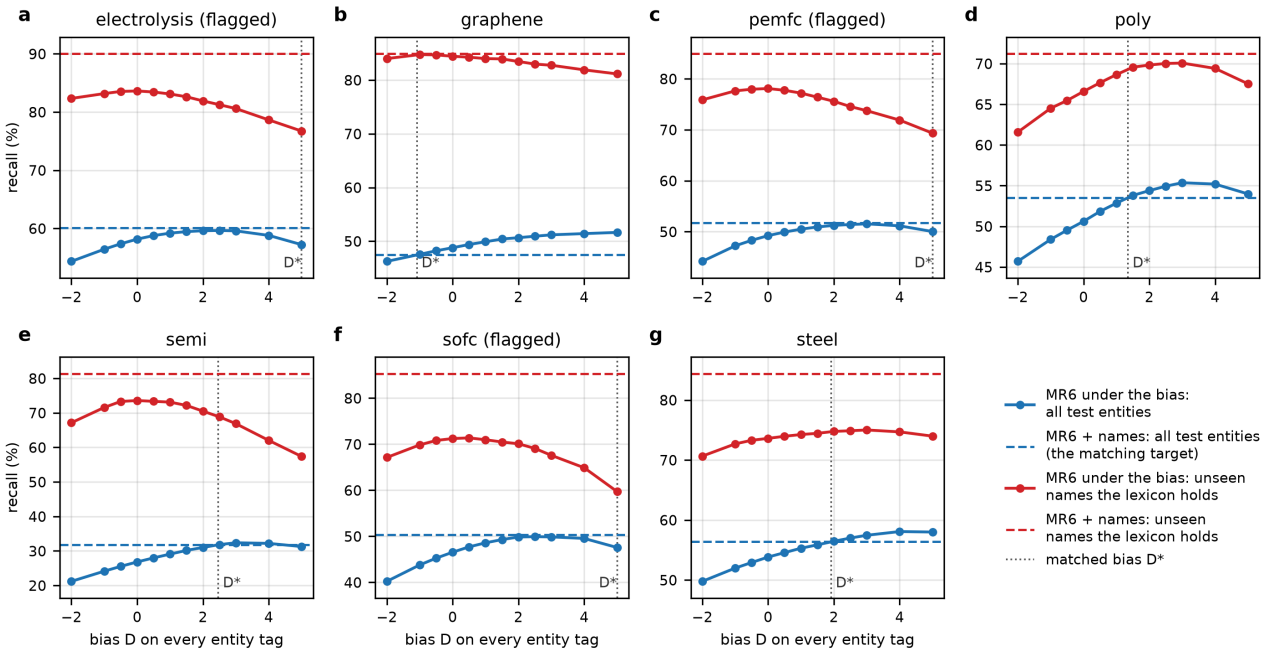


Figure 4. The recall-matched control in each MuLMS subfield. a–g, electrolysis, graphene, pemfc, poly, semi, sofc and steel (decision seeds). Blue lines with dots: MR6's recall over all test entities at each bias D of the grid (the matching, pre-specified); red lines with dots: its recall on the unseen test names the lexicon holds (post hoc); both pooled over the seeds. Dashed lines of the same colours: MR6 + names' values; dotted grey line: the matched bias D*. In the three flagged subfields (**a**, **c** and **f**) the blue curve peaks below its dashed line; in every subfield the red curve stays below its dashed line. Pooled rates carry no error bars.

Tables

Table 1. The three corpora: documents and sentences, annotation, the held-out unit with its test size, and the training set per unit.

Corpus	Documents, sentences	Annotation	Held-out unit (test size)	Training per unit
MaterioMiner [1], release v1.0.1, CC BY 4.0	4 papers on fatigue of metals (s11340, ncomms7164, s41467, met10111458: 41, 81, 201, 153 sentences)	the classes of the dataset's materials mechanics ontology (MMO) at two levels: fine-grained, its 27 classes with most entities, the other classes' entities relabelled O (1,328 entities; e.g. Alloy, Value, Mechanism, StateDescriptor, FatigueTestSpecimen, GrainBoundary); coarse-grained, 27 parent types (2,194 entities)	one paper (41–201 sentences)	the other three papers (275–435 sentences, of which 15% validate: 234–370 training sentences)
MuLMS [3], Hugging Face `timo-pierre-schrader/MuLMS` at revision bc1c97d, CC BY-SA 4.0	50 articles from 7 materials subfields, 10,184 sentences: water electrolysis (electrolysis; 4 articles, 992 sentences), graphene (2, 351), polymer-electrolyte-membrane fuel cells (pemfc; 8, 1,638), polymers (poly; 11, 2,384), semiconductors (semi; 11, 2,040), solid oxide fuel cells (sofc; 4, 928), steel (10, 1,851)	13 types (MAT, VALUE, PROPERTY, FORM, MEASUREMENT, CITE, SAMPLE, TECHNIQUE, DEV, INSTRUMENT, NUM, UNIT, RANGE); we keep the outermost of its nested entities, 31,952 of 46,584	one subfield, every sentence of its articles (351–2,384)	400 sentences of the other subfields' articles, taken in a shuffled order of those articles until 400 are reached (two or three articles, the last cut; a different set for each held-out subfield); 340 train and 60 validate, MaterioMiner's size by design
SOFC-Exp [2], `boschresearch/sofc-exp_textmining_resources`, CC BY 4.0	45 articles on solid oxide fuel cells; 876 of their 9,466 sentences carry entity annotation (the ones that describe experiments)	4 types: MATERIAL, VALUE, DEVICE, EXPERIMENT	9 articles, one of 5 folds (137–248 sentences)	the other 36 articles (534–628 training and 94–111 validation sentences)

Table 2. The twelve pre-registered decisions of the names route (D9 to D12 the follow-ups) in the order taken: corpus and units (with MuLMS's seeds), comparison, the reference's and the system's mean micro F1 (for D11, recall), the mean paired difference with its 95% CI, the two-sided P value of the same t test (computed for reporting; Methods, *Statistics and reproducibility*), and the verdict.

	Corpus, units (MuLMS seeds)	Comparison	Reference	System	Δ [95% CI]	P	Verdict
D1	MuLMS, 7 subfields (300–900)	MR6 + names vs CRF head	48.54	51.73	+3.19 [−1.13, +7.51]	0.12	no
D2	MuLMS, 7 subfields (300–900)	MR6 + names vs MR3	50.81	51.73	+0.92 [−1.63, +3.48]	0.41	no
D3	MaterioMiner fine, 4 papers	MR6 + names vs CRF head	46.46	50.72	+4.27 [+1.39, +7.15]	0.018	passes
D4	MaterioMiner coarse, 4 papers	MR6 + names vs CRF head	58.60	60.38	+1.77 [+0.67, +2.88]	0.014	passes
D5	pooled: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 + names vs CRF head			+3.58 [+0.95, +6.21], neither corpus below 0	0.013	passes
D5′	the same with MuLMS seeds 300–900 (the protocol's batch; post hoc reading)				+3.58 [+1.04, +6.13]	0.011	above 0 (post hoc)
D6	SOFC-Exp, 5 article folds	MR6 + names vs CRF head	82.79	82.75	−0.04 [−1.21, +1.14]	0.94	no
D7	pooled, 11 units (300–900)	MR6 + names vs MR6			+0.80 [−0.66, +2.26]	0.25	no
D8	MuLMS, 7 subfields (300–900)	own-topic corpus vs mixed corpus	51.91	51.73	−0.18 [−2.01, +1.65]	0.81	no
D9	MaterioMiner fine, 4 papers (follow-up)	multi-task + MR6 + names vs multi-task + MR6	52.99	53.95	+0.96 [−0.01, +1.92]	0.052	no
D10	MaterioMiner coarse, 4 papers (follow-up)	multi-task + MR6 + names vs multi-task + MR6	60.40	61.25	+0.85 [−0.72, +2.42]	0.18	no
D11	MuLMS, 7 subfields (300–900; follow-up)	MR6 + names vs MR6 at the recall-matched bias (3 of 7 subfields at the grid's end): recall on the unseen names the lexicon holds	71.92	83.16	+11.24 [+3.31, +19.18]	0.013	passes (as registered)
D12	MuLMS, 7 subfields (300–900; follow-up)	MR6 + names vs MR6 at the recall-matched bias (3 of 7 subfields at the grid's end): micro F1	48.47	51.73	+3.26 [−0.79, +7.31]	0.096	no

The reference F1 on MuLMS is the 7-seed mean (48.54); the 10-seed mean quoted in the Introduction is 48.66. D5 and D5′ are the two readings of one protocol text (Methods, *Evaluation protocol*). Every column is micro F1 except D11's. For D11 and D12 the reference is MR6 decoded at the bias that matches the names' overall recall; in three subfields no bias on the grid (−2 to 5) reaches it, and the protocol matched them at its end, D = 5, where MR6's recall is 1.7 to 2.9 points lower. D11's columns are recall on the unseen test names the lexicon holds, the mean over the subfields of rates pooled over the seeds (Results, *What the names do*). Without the three flagged subfields (pre-specified), D11 is +6.05 [−3.33, +15.44] and D12 −0.03 [−2.78, +2.72]. D5's criterion was adopted on 20 October, when mention replacement without names was already known to meet it (Methods, *Evaluation protocol*).

Table 3. Every comparison among the four systems: over the 11 held-out units on both MuLMS seed bases (top) and over MaterioMiner's four papers per granularity (bottom); the mean paired difference of micro F1 with its 95% CI, for the 11 units the units above 0 and the exact two-sided sign-flip P value, and the status of each entry.

Comparison, 11 units	MuLMS seeds 300–900: Δ micro F1 [95% CI]	units above 0	sign-flip P	MuLMS seeds 0–900: Δ micro F1 [95% CI]	units above 0	sign-flip P	status
MR3 − CRF head	+2.46 [+1.00, +3.91]	11/11	0.001	+2.35 [+0.94, +3.75]	10/11	0.003	post hoc
MR6 − MR3	+0.33 [−0.25, +0.92]	8/11	0.23	+0.22 [−0.30, +0.74]	8/11	0.37	post hoc (per corpus pre-specified, 29 October)
MR6 − CRF head	+2.79 [+1.20, +4.38]	10/11	0.002	+2.56 [+0.94, +4.18]	10/11	0.005	post hoc (per corpus pre-specified, 29 October)
MR6 + names − MR6	+0.80 [−0.66, +2.26]	7/11	0.24	+1.02 [−0.56, +2.59]	6/11	0.18	D7 on seeds 300–900; post hoc on 0–900
MR6 + names − MR3	+1.13 [−0.51, +2.76]	7/11	0.16	+1.23 [−0.22, +2.68]	8/11	0.09	post hoc (on MuLMS alone, D2)
MR6 + names − CRF head	+3.58 [+1.04, +6.13]	10/11	0.007	+3.58 [+0.95, +6.21]	9/11	0.011	D5 on seeds 0–900; D5′ a post hoc reading

MaterioMiner, 4 held-out papers (10 seeds)	fine-grained [95% CI]	coarse-grained [95% CI]	status
MR3 − CRF head	+2.77 [+1.51, +4.03]	+1.28 [+0.06, +2.51]	companion decisions (protocol of 16 October, 14:39), both passing
MR6 − MR3	+0.20 [−1.47, +1.88]	−0.21 [−2.71, +2.29]	pre-specified, 29 October
MR6 − CRF head	+2.98 [+0.09, +5.86]	+1.07 [−1.82, +3.97]	pre-specified, 29 October
MR6 + names − MR6	+1.29 [−2.51, +5.10]	+0.70 [−1.60, +3.00]	pre-specified, 29 October (fine-grained: D7's per-corpus condition)
MR6 + names − MR3	+1.50 [−1.76, +4.75]	+0.49 [+0.06, +0.92]	pre-specified, 20 October (00:51)
MR6 + names − CRF head	+4.27 [+1.39, +7.15]	+1.77 [+0.67, +2.88]	D3; D4

Table 4. Recall, precision and F1 on the unseen test names, pooled over seeds and MuLMS's seven held-out subfields (post hoc).

MuLMS, unseen names (88% of the test entities); 70 runs per row (seeds 0–900), 49 for the mixed corpus (seeds 300–900)	recall	precision	F1
CRF head	39.4	55.8	46.15
MR3	43.1	54.3	48.10
MR6	43.7	53.9	48.28
MR6 + names, the topic corpus	47.3	50.9	49.01
MR6 + names, a mixed corpus	46.7	52.4	49.35

On the decision seeds alone (49 runs per row) the topic corpus's row reads 46.9 / 51.3 / 49.0, MR6's 43.9 / 54.2 / 48.5 and the CRF head's 39.1 / 55.8 / 46.0, so the two corpora's rows compare on the same seeds.

Table 5. Recall gains on the unseen test names by lexicon membership (post hoc): per comparison, the gain on the names the lexicon holds and on the others, their difference, paired over the units with 95% CIs, and the units where the gain on the held names is the larger. The lexicon holds 22.2% of MuLMS's unseen test names and 9.8% and 18.2% of MaterioMiner's (fine- and coarse-grained). The mixed-corpus row is split by the topic corpus's lexicon, which was the one rebuilt. MuLMS on the decision seeds, MaterioMiner on 10 seeds.

setting	comparison	recall gain, names in the lexicon [95% CI]	recall gain, names not in it [95% CI]	in − out [95% CI]	units with in > out
MuLMS, 7 subfields	MR6 + names − MR6	+7.29 [+3.27, +11.30]	+1.12 [−0.78, +3.02]	+6.17 [+2.44, +9.90]	7/7
MuLMS, 7 subfields	MR6 − CRF head	+3.39 [+0.92, +5.85]	+5.22 [+1.92, +8.52]	−1.84 [−5.27, +1.60]	3/7
MuLMS, 7 subfields	MR6 + names (mixed corpus) − MR6	+6.16 [+1.32, +11.01]	+1.31 [−1.07, +3.68]	+4.86 [+1.50, +8.21]	7/7
MaterioMiner fine, 4 papers	MR6 + names − MR6	+4.62 [−5.27, +14.50]	+0.29 [−4.23, +4.81]	+4.33 [−5.33, +13.98]	3/4
MaterioMiner coarse, 4 papers	MR6 + names − MR6	+0.36 [−1.11, +1.84]	+1.91 [−1.46, +5.28]	−1.54 [−3.60, +0.51]	0/4

Table 6. The names against the recall-matched control on MuLMS's held-out subfields (follow-up; seeds 300–900): the names' recall over all test entities pooled over the seeds, MR6's at bias 0, the matched bias D*; recall on the unseen test names the lexicon holds for the names, MR6 at D = 0 and the control at D*; and the names minus the control at D* in that recall (D11), micro F1 (D12) and precision (pre-specified), with the means over the subfields and their 95% CIs. In a flagged subfield the names' recall exceeds MR6's at every bias of the grid; it is matched at the grid's end, as the protocol fixed, and in parentheses at the grid value of the nearest recall (post hoc). The last two rows compare the names with MR6 at D = 0 (the recall post hoc, the F1 D7's MuLMS part) and with MR6 at whichever bias gives it its highest recall on the held names (post hoc).

Subfield	names' recall	MR6's recall at D = 0	D*	recall on the held names: names / MR6 at 0 / control at D*	Δ recall, held names	Δ micro F1	Δ precision
electrolysis	60.13	58.17	5 (flagged; 2.5)	90.02 / 83.62 / 76.73	+13.29 (+8.76)	+7.13 (+2.30)	+10.28 (+3.75)
graphene	47.45	48.79	−1.08	84.91 / 84.43 / 84.68	+0.22	−1.31	−3.05
pemfc	51.74	49.23	5 (flagged; 3)	84.92 / 78.09 / 69.33	+15.59 (+11.18)	+6.43 (+3.28)	+11.03 (+6.50)
poly	53.52	50.61	1.35	71.22 / 66.58 / 69.29	+1.93	−1.48	−2.55
semi	31.78	26.82	2.47	81.36 / 73.58 / 68.96	+12.40	+2.20	+6.90
sofc	50.36	46.55	5 (flagged; 2.5)	85.32 / 71.23 / 59.72	+25.61 (+16.23)	+9.39 (+4.15)	+16.93 (+8.82)
steel	56.37	53.83	1.92	84.39 / 73.61 / 74.72	+9.67	+0.47	+0.93
mean, 7 subfields					+11.24 [+3.31, +19.18] (D11, passes)	+3.26 [−0.79, +7.31] (D12, no)	+5.78 [−1.22, +12.79]
mean without the flagged, 4 subfields (pre-specified)					+6.05 [−3.33, +15.44]	−0.03 [−2.78, +2.72]	+0.56 [−6.74, +7.85]
mean, the flagged at the nearest recall, 7 subfields (post hoc)					+8.63 [+3.35, +13.90]	+1.37 [−0.66, +3.41]	+3.04 [−1.32, +7.41]
names − MR6 at D = 0, 7 subfields (recall post hoc; F1 D7's MuLMS part)					+7.29 [+3.27, +11.30]	+0.51 [−1.51, +2.53]	
names − MR6 at its best bias for the held names, 7 subfields (post hoc)					+6.52 [+2.15, +10.89]		